# CAP182 SS2 - Part A: Related Work and Data Description


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| **Module** | CAP182 Capstone Project |
| **Project** | STADIOEquities account-activation proof of viability |


## Publication 1

| Field | Detail |
|---|---|
| **Title** | Data driven methodology for predicting successful bank telemarketing |
| **Harvard reference** | Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', *Decision Support Systems*, 62, pp. 22-31. https://doi.org/10.1016/j.dss.2014.03.001. |
| **Link** | https://doi.org/10.1016/j.dss.2014.03.001 |
| **Modelling** | Compared Logistic Regression, Decision Trees, a Neural Network and a Support Vector Machine. An automatic/semi-automatic selection method was used to reduce 150 candidate inputs down to 22. Models were assessed using ROC-AUC and area under the cumulative lift curve based on a temporal rolling window approach. The Neural Network proved superior (AUC = 0.80; ALIFT = 0.70) and selected 79% of the customer base, by picking the top half from rankings. |
| **Preprocessing techniques** | Combined business logic with forward selection of features; reduced 150 potential client, product and socio-economic inputs to 22; isolated training data from evaluation period post-July 2012 and used rolling windows. |
| **Datasets used** | A set of 52,944 real world telemarketing calls recorded from a Portuguese retail bank from 2008 to 2013. UCI bank-additional-full dataset is a subset of the dataset comprising 41,188 highly relevant calls from May 2008 through November 2010. |
| **Key takeaways** | (1) Realistic AUC of about 0.80 is a benchmark for this financial conversion problem. (2) Rolling window evaluation illustrates that validation of a model should be done in temporal order rather than randomly splitting data. (3) The advantage of the Neural Network over Logistic Regression justifies comparison between transparent baseline and non-linear model and evaluation of ranking capabilities as outreach capacity. |

# Publication 2

| Field | Detail |
|---|---|
| **Title** | Bayesian regression analysis to predict the subscription of bank term deposits |
| **Harvard reference** | Tanvir, M.F., Hossain, M.M. and Jishan, M.A. (2024) 'Bayesian Regression for Predicting Subscription to Bank Term Deposits in Direct Marketing Campaigns', *arXiv*, arXiv:2410.21539. Available at: https://arxiv.org/abs/2410.21539 (Accessed: 1 September 2026). |
| **Link** | https://arxiv.org/abs/2410.21539 |
| **Modelling** | Conducted Fitted Bayesian logistic and probit regressions with MCMC posterior estimation. Leave-One-Out Cross-Validation greatly preferred the logit model: the expected difference in log pointwise predictive density between the logit model and probit was -1,860.9 (SE 49.7). |
| **Preprocessing techniques** | Randomly chose 10,000 samples from the 41,188-rows data set and oversampled to get 5,000 positive and 5,000 negative examples for computationally tractable balanced modeling. |
| **Datasets used** | UCI Bank Marketing bank-additional-full data: 41,188 Portuguese bank direct-marketing campaigns with 21 attributes, including binary target variable. This is the same public dataset used in the current project. |
| **Key takeaways** | (1) Performance of Logit on the same problem implies the relevance of Logistic Regression to be used as baseline. (2) The specific balancing procedure proves that class imbalance should be considered and this project takes class-weighted approach while preserving natural distribution in test. (3) This paper incorporates call duration and indicates high uncertainty in the sample predictions, while an operational STADIOEquities model has to ignore post-fact information and analyze multiple metrics of discrimination and minority class. |

## Publication 3

| Field | Detail |
|---|---|
| **Title** | Addressing the class imbalance problem in churn prediction using ensembles and SMOTE |
| **Harvard reference** | Suguna, R., Suriya Prakash, J., Aditya Pai, H., Mahesh, T.R., Vinoth Kumar, V. and Yimer, T.E. (2025) 'Mitigating class imbalance in churn prediction with ensemble methods and SMOTE', *Scientific Reports*, 15, article 16256. https://doi.org/10.1038/s41598-025-01031-0. |
| **Link** | https://doi.org/10.1038/s41598-025-01031-0 |
| **Modelling** | Nine individual classifiers and six homogeneous ensembles were evaluated: Logistic Regression, KNN, decision trees, SVMs, MLP, Random Forest, Gradient Boosting, AdaBoost, XGBoost, LightGBM and CatBoost. The models were optimised using grid search. Among the boosting ensembles, AdaBoost and Gradient Boosting were the best-performing after applying SMOTE; according to the article, AdaBoost yielded the highest F1 score of around 0.876. |
| **Preprocessing techniques** | RowNumber, CustomerId and Surname features were dropped; Geography and Gender were one-hot encoded; all numeric features were standardised; train/test was done on an 80/20 split basis; SMOTE resampling was done only to the training set; grid search was used to tune chosen models. |
| **Datasets used** | Churn Modelling – Kaggle: dataset of bank customers, 10,000 examples, 14 features and an Exit target with around 20% churn and 80% non-churn cases. |
| **Key takeaways** | ((1) While accuracy can hide low performance on the minority class, balanced accuracy, recall, F1 and PR-AUC should also be reported. (2) Any SMOTE or other resampling should be used to the training data and compared to class weighting rather than being assumed better. (3) Tuning and improvements from ensembling are important for XGBoost; however, different churn target, dataset and resampling make the results of the paper not comparable. |

## Public dataset for Part B

| Field | Detail |
|---|---|
| **Name** | Bank marketing (bank-additional-full) |
| **Harvard reference** | Moro, S., Rita, P. and Cortez, P. (2014) *Bank Marketing* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5K306. |
| **Link** | https://archive.ics.uci.edu/dataset/222/bank+marketing |
| **Number of instances** | 41,188 instances in bank-additional-full.csv. UCI provides a 4,119 instance 10% sample dataset and another older version containing 45,211 instances named bank-full with a corresponding 4,521 instance sample dataset. This project will utilize the full 41,188 instance dataset and construct its own 80/20 holdout set. |
| **Number of feature columns** | 20 input features plus target y. Excluding duration as leakage, there are 19 raw model inputs prior to feature engineering. |
| **Number of classes** | Two possible outcomes: yes (subscribed) and no (not subscribed). There are 4,640 instances belonging to positive class and 36,548 instances belonging to negative class, therefore positive ratio is 11.3%. |
| **Target range** | Not applicable: it is binary classification problem, not regression. |
| **Origin** | Actual records for real-world phone-based marketing campaign carried out by a Portuguese retail bank between May 2008 and November 2010. The target variable tells if the contacted individual subscribed to term deposit or not. |

## Relevance to STADIOEquities

The problem for SS1 consists of determining if the newly registered user will perform his/her first deposit within 30 days. In the case of the public problem, it involves determining if the contacted bank customer will sign up for a term deposit. These two are imbalanced binary predictions of a significant financial commitment on behalf of the customer. Both models need a way to prioritize customers for intervention while limiting the expense of false positives.

The comparison is not exact in that telemarketing activity and macroeconomic variables are not the substitutes of STADIOEquities KYC progress, application behavior, acquisition channel, or support history. The public data is, however, used to demonstrate the entire process flow, rather than suggest that this model is ready for implementation. The main transferable learnings from this project involve: leakage elimination, preprocessing in training only, dealing with imbalance, comparing a linear baseline against non-linear ensemble, and using threshold dependent and ranking metrics for evaluation.

## References

Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', *Decision Support Systems*, 62, pp. 22-31. https://doi.org/10.1016/j.dss.2014.03.001.

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Suguna, R., Suriya Prakash, J., Aditya Pai, H., Mahesh, T.R., Vinoth Kumar, V. and Yimer, T.E. (2025) 'Mitigating class imbalance in churn prediction with ensemble methods and SMOTE', *Scientific Reports*, 15, article 16256. https://doi.org/10.1038/s41598-025-01031-0.

Tanvir, M.F., Hossain, M.M. and Jishan, M.A. (2024) 'Bayesian Regression for Predicting Subscription to Bank Term Deposits in Direct Marketing Campaigns', *arXiv*, arXiv:2410.21539. Available at: https://arxiv.org/abs/2410.21539 (Accessed: 1 September 2026).